

19 ANS: D

$$E = Q / 4\pi\epsilon_0 r^2$$

Distance from midpoint to each charge is 0.50 m

Electric field strength due to one point charge,

$$E = Q / 4\pi\epsilon_0 r^2 = (2.0 \times 10^{-6}) / 4\pi(8.85 \times 10^{-12})(0.50)^2$$

$$E = 7.19 \times 10^4 \text{ Vm}^{-1}$$

Electric field due to both charges is double the value above because the charges are unlike. $E = 1.4 \times 10^5 \text{ Vm}^{-1}$